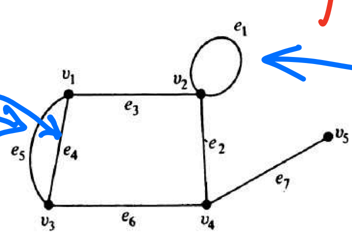
Graph Theory Work-along Sheet

Name: ______________________________; **Roll Number:** ______________
Lecture Batch: ____________________ ; **Lab Batch:** ______________________

A __linear__ graph (or simply a __graph__) G = (V, E) consists of a set of objects V = {v_1, v_2, . . .} called __vertices__, and another set E = {e_1, e_2,. . .}, whose elements are called __edges__, such that each edge e_k is identified with an __unordered__ pair (v_i, v_j) of vertices. The vertices v_i and v_j associated with edge e_k are called the end vertices of e_k. The most common representation of a graph is by means of a __diagram__, in which the __vertices__ are represented as points and each __edge__ as a line segment joining its end vertices. Often, this diagram itself is referred to as the __graph__.

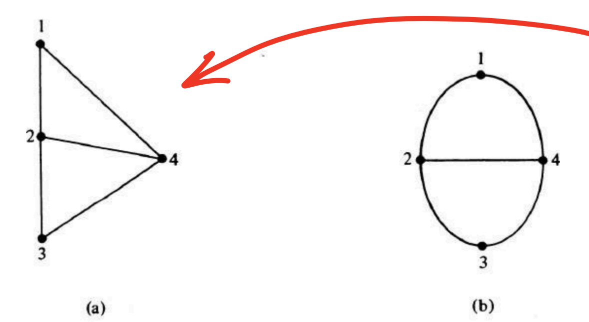
The edge having the same vertex as both its end vertices is called a __self-loop__. If more than one edge is associated with a given pair of vertices, then such edges are referred to as __parallel__ edges.

Parallel edges



Self-loop

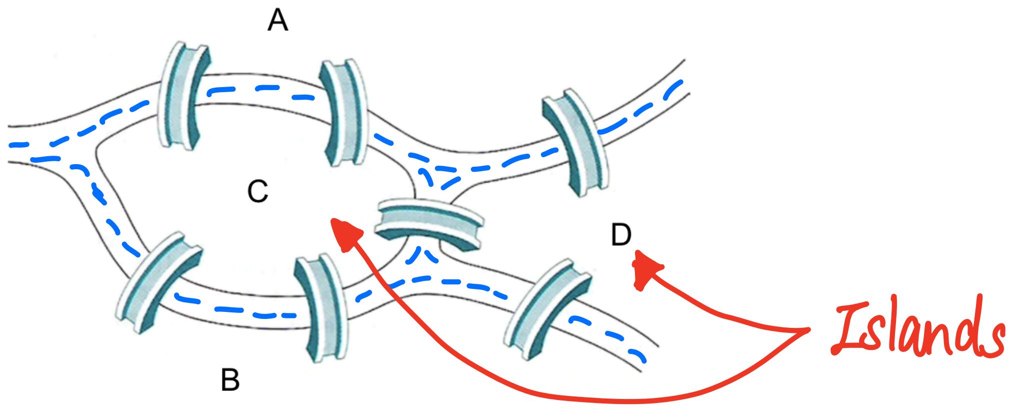
A graph that has neither self-loops nor parallel edges is called a __simple__ graph.



Same graph

Question: Are the above two graphs the same or different?
Answer: __Same__ because incidence between edges and vertices is the same in both cases.

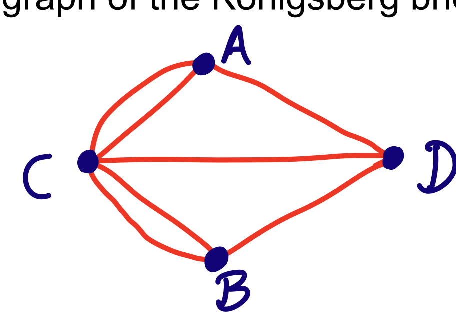
A graph can be used to represent almost any physical situation involving discrete objects and their relationships.



Seven Bridges of Königsberg problem:

- **Location:** The problem is set in Königsberg (now Kaliningrad) on the Pregel River.
- **Geography:** It involves four land areas: two islands (C and D) and two riverbanks (A and B).
- **Connections:** These four land areas are connected by a total of seven bridges.
- **The Challenge:** To find a path that starts at any land area, crosses every bridge exactly once, and returns to the starting point.
  Euler's Method: Leonhard Euler modelled the city as a mathematical graph:
  - **Vertices (Points):** Represent the land areas (A, B, C, D).
  - **Edges (Lines):** Represent the bridges.

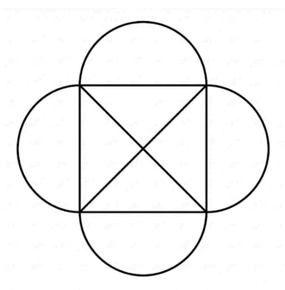
  Try to draw the graph of the Königsberg bridge problem.

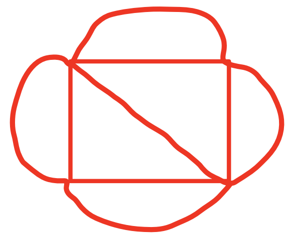


- **The Solution:** Euler proved that such a path is [possible/impossible] to create. How? Think about it. 🤔

Childhood Activity:

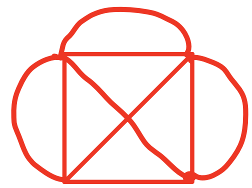
Remember the following childhood activity, where we had to draw the following figure without lifting the pen from the paper and without retracing a line.





[Your first attempt]

failed!!!



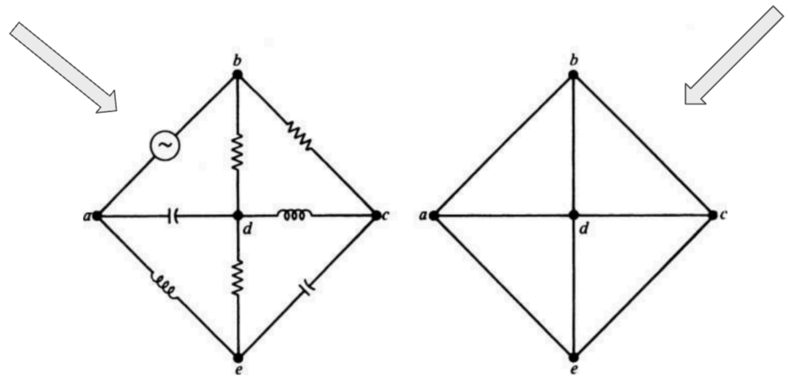
[Your second attempt]

failed!!!

Is it possible to do it? [YES / NO]

Electrical network

Graph of Electrical network



A graph with a finite number of vertices as well as a finite number of edges is called a finite graph; otherwise, it is an infinite graph.

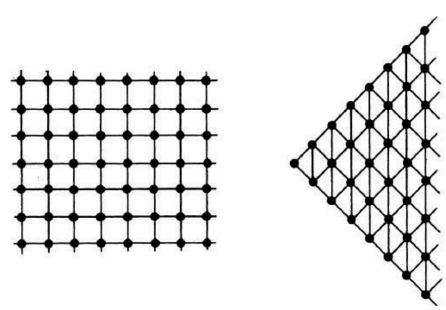
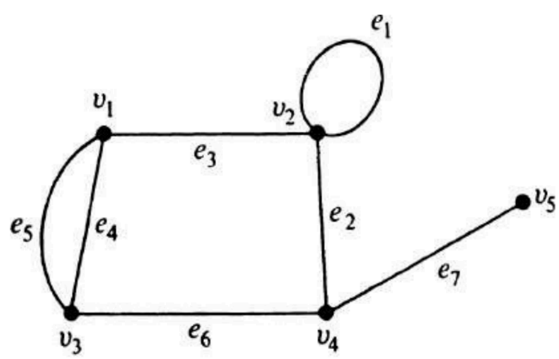


Fig: Portions of two infinite graphs

In this discussion, we shall confine ourselves to the study of finite graphs, and unless otherwise stated, the term “graph” will always mean a finite graph.

The number of edges incident on a vertex v_i, with self-loops counted twice, is called the degree, $d(v_i)$, of vertex v_i. The degree of a vertex is sometimes also referred to as its __valency__.

Find the degree of all the vertices of the following graph:



Degree Sequence

$\{4, 3, 3, 3, 1\}$

$d(v_1)$= __3__, $d(v_2)$= __4__, $d(v_3)$= __3__, $d(v_4)$= __3__, $d(v_5)$= __1__

Now, add all the degrees.

$d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5)$

= __3__ + __4__ + __3__ + __3__ + __1__

= __14__

Is there any relation between the sum of the degrees of all the vertices and the number of edges? State your observations.

There are 7 edges (e_1 to e_7) & sum of degrees of all the vertices is 14, i.e., twice the number of edges.

Handshaking Lemma:

The Handshaking Lemma states that the __sum__ of the degrees of all the vertices in any finite graph must equal twice the number of __edges__.

This is true because every edge connects exactly two vertices, meaning it contributes exactly 2 to the total sum of the degrees.

Handshaking Lemma

$$\boxed{\sum_{i=1}^{N} d(v_i) = 2e}$$

$$\sum \deg(v_i) = 2|E|$$

Now, write the degrees of the vertices of the earlier-mentioned graph in non-increasing order.

$4 \geq 3 \geq 3 \geq 3 \geq 1$

$\{4, 3, 3, 3, 1\}$

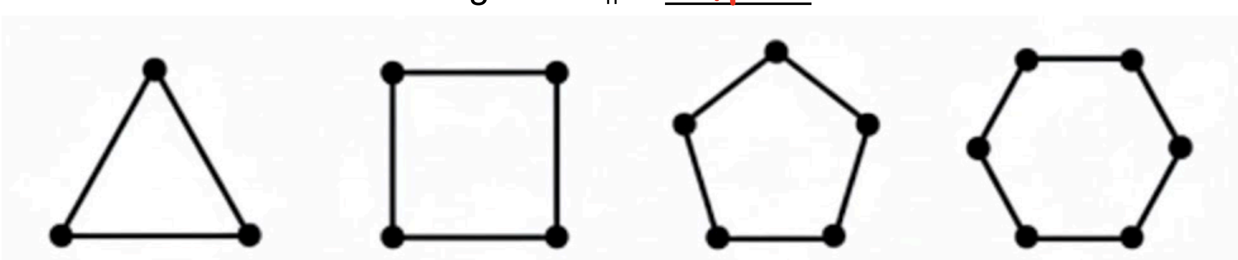
This degree sequence provides a fundamental summary of a graph's structure.

Degree Sequence:

The degree sequence of a graph is simply a list of the degrees of all its vertices (nodes), typically arranged in non-increasing order.

Cycle Graph:

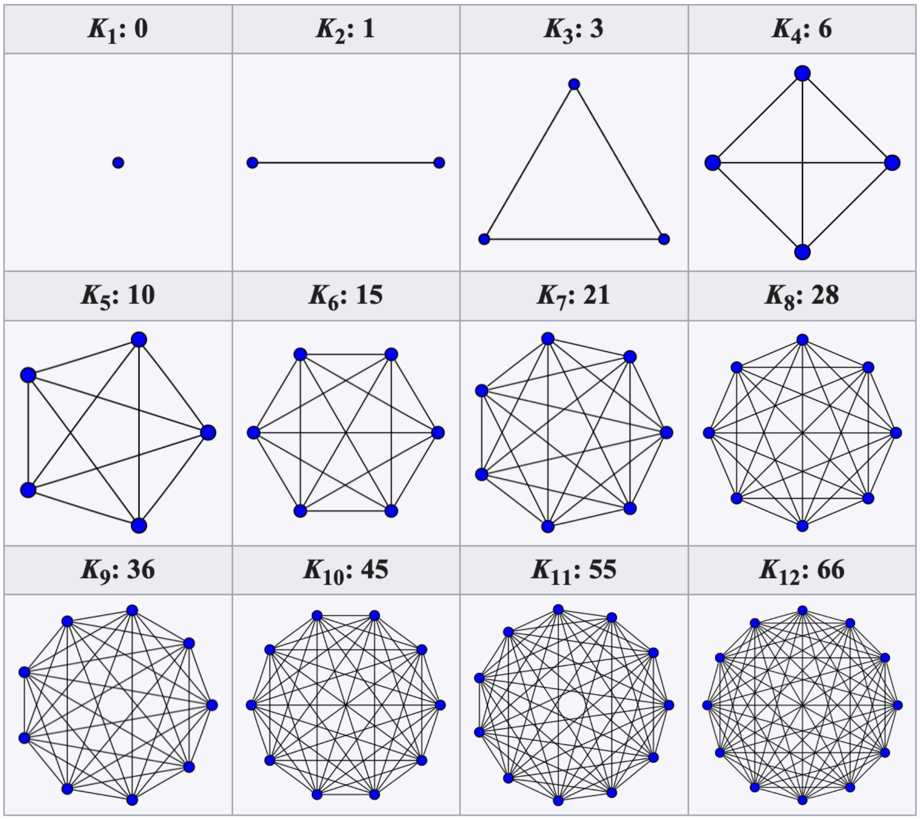
- Cycles are simple graphs with n ≥ 3 vertices and edges: {1, 2}, {2, 3}, ..., {n−1, n}, and {n, 1}.
- A cycle with n vertices is denoted as C_n.
- The total number of edges in C_n is n.



C_3 C_4 C_5 C_6

Complete Graph:

- A simple graph of n vertices having exactly one edge between each pair of vertices is called a complete graph.
- A complete graph of n vertices is denoted by K_n.

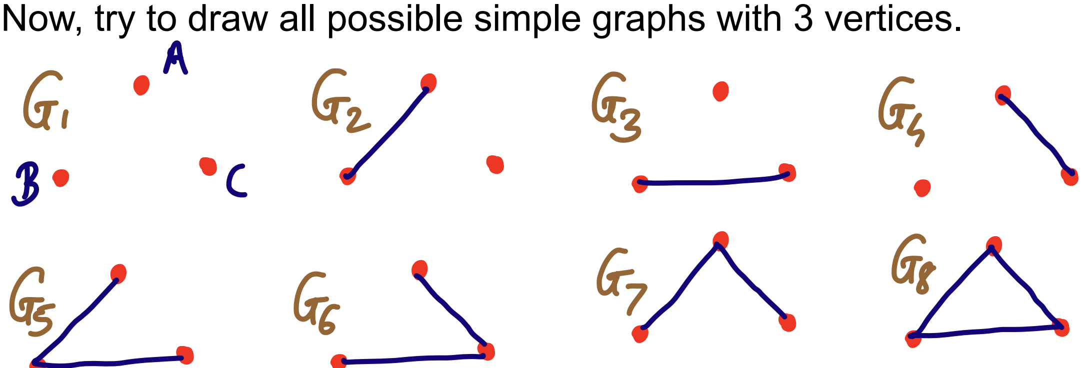


- The total number of edges in K_n is ${}^{n}C_2$ = $\frac{n!}{2!(n-2)!}$ = $\frac{n(n-1)}{2}$.

Now, try to draw all possible simple graphs with 2 vertices.

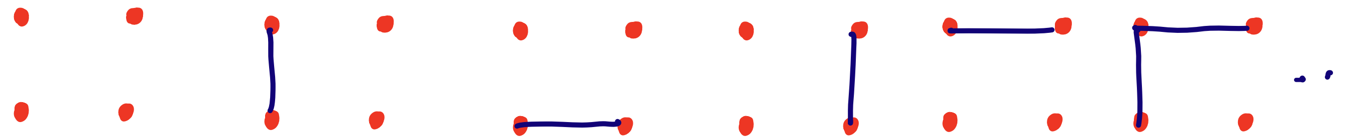
Number of simple graphs with 2 vertices = 2.

Now, try to draw all possible simple graphs with 3 vertices.



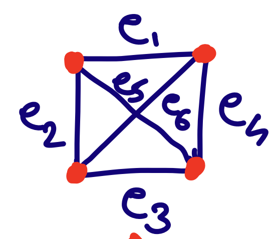
Number of simple graphs with 3 vertices = 8.

Now, try to draw all possible simple graphs with 4 vertices.



Time consuming

Consider complete graph with 4 vertices, i.e., K_4. $K_4 \Rightarrow$

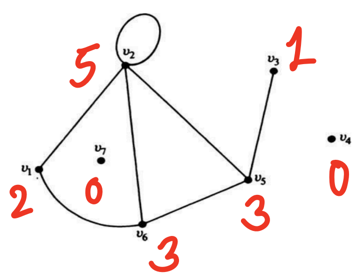
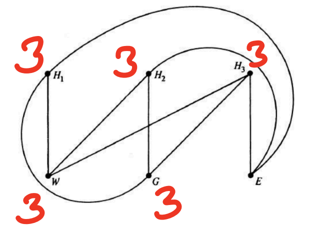
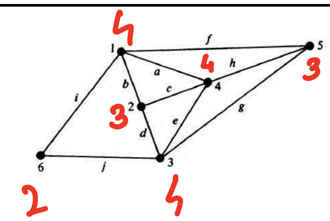


Number of simple graphs with 4 vertices = 2^6.

present absent

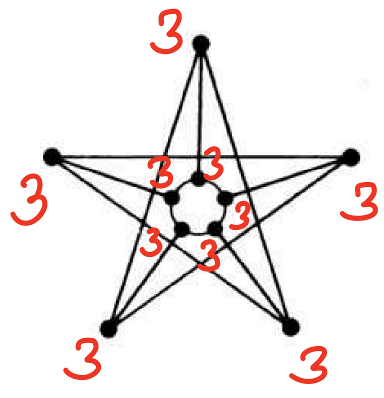
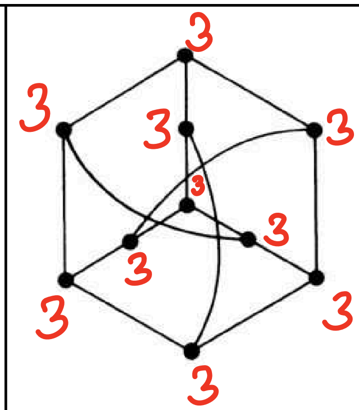
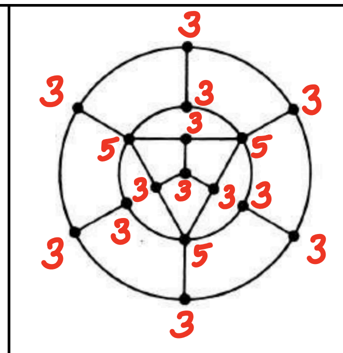
Observe the pattern. Think about the number of simple graphs with n vertices. 🤔
Hint: There are 2 choices (include and exclude) for every edge of a complete graph.

	n = 3	n = 4	n
Maximum degree of any vertex in a simple graph with n vertices	$3-1=\underline{\underline{2}}$	$4-1=\underline{\underline{3}}$	$\underline{\underline{n-1}}$
Maximum number of edges in a simple graph with n vertices	${}^3C_2=\underline{\underline{3}}$	${}^4C_2=\underline{\underline{6}}$	$\underline{\underline{{}^nC_2}}$
Maximum number of simple graphs possible with n vertices	$2^{{}^3C_2}=2^3=\underline{\underline{8}}$	$2^{{}^4C_2}=2^6=\underline{\underline{64}}$	$\underline{\underline{2^{{}^nC_2}}}$

			
Number of Vertices	7	6	6
Number of Edges	7	9	10
Simple graph / Multi graph	Multi graph	Simple graph	Simple graph
Number of pendant (degree-1) vertices	1	0	0
Number of isolated (degree-0) vertices	2	0	0
Is the graph complete?	No	No	No
Is the graph regular?	No	Yes	No
Degree sequence	{5,3,3,2,1,0,0}	{3,3,3,3,3,3}	{4,4,4,3,3,2}

Regular graph is a graph in which all vertices have same degrees

Swap & Check Contest

Marks		10 marks	10 marks	10 marks
				
1	Number of Vertices	10	10	16
1	Number of Edges	15	15	27
1	Simple graph / Multi graph	Simple graph	Simple graph	Simple graph
1	Number of pendant vertices	0	0	0
1	Number of isolated vertices	0	0	0
1	Is the graph complete?	No	No	No
1	Is the graph regular?	Yes	Yes	No
3	Degree sequence	{3,3,3,3,3,3,3,3,3,3}	{3,3,3,3,3,3,3,3,3,3}	{5,5,5,3,3,3,3,3,3,3,3,3,3,3,3,3}